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Fitness-for-Service Assessment - 6-RA-103-11 Local Metal Loss (Level 2)

Scope and Objectives

This assessment addresses the structural integrity of 6-RA-103, a rich amine header located within the ATU-100 at the Hamilton Plant. The evaluation focuses specifically on 6-RA-103-11, where recent ultrasonic data indicated significant wall thinning. The objective is to determine the fitness-for-service of the component under design conditions of 862 kPa(g) and 82 °C using the methodologies defined in API 579-1/ASME FFS-1.

Priya Natarajan, acting as the FFS engineer, is responsible for the engineering calculations and the determination of the remaining strength factor. The scope is limited to the local metal loss identified at the specified CML and assumes that the upstream let-down valve trim is restored to prevent further accelerated erosion-corrosion. All calculations utilize the design requirements of ASME B31.3 for the A106 Gr.B material.

Inspection Data and Observations

The primary data source for this assessment is LNDE-RPT-20241202-036-R0, which provides detailed PAUT corrosion mapping of the affected area. The report, dated 2024-12-10, confirms a minimum wall thickness of 5.82 mm at 6-RA-103-11. This represents a significant reduction from the nominal thickness of 10.97 mm and is approaching the established t-min of 4.57 mm.

Historical data for 6-RA-103 shows a stable corrosion rate of approximately 0.234 mm/y between 2014 and 2019. However, the latest survey results indicate a sharp increase in the short-term corrosion rate to 0.600 mm/y. This acceleration is attributed to localized turbulence and flow-accelerated corrosion, necessitating a Level 2 assessment to validate continued operation.

Fitness-for-Service Evaluation

The Level 2 assessment performed by Priya Natarajan followed the Part 5 procedures of API 579-1/ASME FFS-1 for local metal loss. The evaluation considered the longitudinal and circumferential extent of the thinning as mapped in LNDE-RPT-20241202-036-R0. Based on the current thickness of 5.82 mm, the component maintains a remaining strength factor above the allowable threshold for the design pressure of 862 kPa(g).

The assessment confirms that 6-RA-103-11 is acceptable for continued operation in its current state. The calculation accounts for the A106 Gr.B material properties and the geometric constraints of the 168.3 mm OD piping. The validity of this conclusion is contingent upon the mitigation of the underlying flow issues associated with the let-down valve.

Remaining Life and Recommendations

Per site rules, the remaining life is calculated using the short-term rate of 0.600 mm/y and the t-min of 4.57 mm. This results in a remaining life of 2.1 years. Given the rich amine service and the recent acceleration in metal loss, the inspection interval must be adjusted to ensure the integrity of the ATU-100 header.

It is recommended to re-measure the thickness at 6-RA-103-11 within 12 months to monitor the effectiveness of the valve trim restoration. This interval is less than half of the calculated remaining life, adhering to the conservative safety margins required for Class 2 amine service. Future inspections should continue to utilize PAUT to track the morphology of the local metal loss.

Sign-off	Name	Function	Date	Signature
Prepared	Priya Natarajan	FFS engineer, P.Eng.	2025-02-24	_____